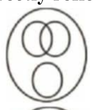
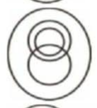
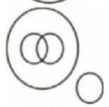
meat. There were some vegetarians who did not accept either. The rest accepted both fish and meat.
Which one of the following logic diagrams correctly reflects the above situation?

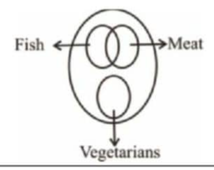
(a) 

(b) 

(c) 

(d) 

Ans. (a) :

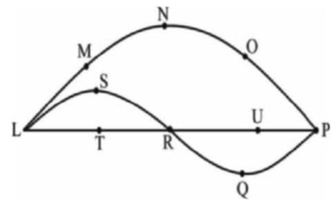


138. A, B, C, D, E, F and G are members of a family consisting of 4 adults and 3 children, two of whom, F and G are girls. A and D are brothers and A is a doctor. E is an engineer married to one of the brothers and has two children. B is married to D and G is their child. Who is C?

(a) G's brother (b) F's father
(c) E's father (d) A's son

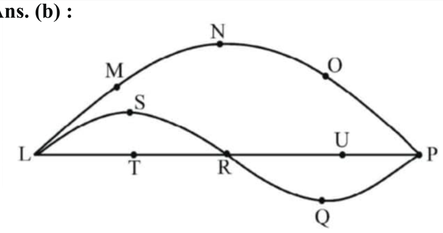
Ans. (d) : E is married to A or D, but B is married to D, so E is married to A.
Now ABED are four adult and C, F, G are three children B and D have child G
A and E have two children they are C & F now only F and G are girls, so C is a boy.
Hence C is A^{rs} and E^{rs} son.

139. LMNOP, is a semicircle with center at R and diameter LP, LSR and RQP are also semicircles with centres at T and U and diameters LR = RP = 1/2 LP. The ratio of perimeter of LMNOP and LSRQP is:



(a) 0.75 : 1 (b) 1 : 1
(c) 1 : 0.75 (d) 1.25 : 1

Ans. (b) :



Let LR = R

Perimeter of LMNOP = $\frac{2\pi R}{2} = \pi R$

Perimeter of LSRQP = Perimeter of LSR + Perimeter of RQP

$= \frac{\pi R}{2} + \frac{\pi R}{2} = \pi R$

$\therefore$ Perimeter $\left(\frac{LMNOP}{LSRQP}\right) = \frac{\pi R}{\pi R} = 1 : 1$

140. A man purchases two clocks A and B at a total cost of Rs. 650. He sells A with 20% profit and B at a loss of 25% and gets the same selling price for both the clocks. What are the purchasing prices of A and B respectively?

(a) Rs. 225; Rs. 425 (b) Rs. 250; Rs. 400
(c) Rs. 275; Rs. 375 (d) Rs. 300; Rs. 350

Ans. (b) : Let the cost price of A be x and that of B be y.

$\therefore x + y = 650$ (Total cost) ……..(i)

For A $\rightarrow$ Profit = 20%

$\therefore 20 = \frac{SP - CP}{CP} \times 100$

$\frac{20}{100} = \frac{SP - x}{x} \Rightarrow \frac{SP}{x} = \frac{1}{5}$

$= SP \text{ of } A = \frac{6x}{5}$

For B $\rightarrow$ Loss = 25%

$\therefore 25 = \frac{CP - SP}{CP} \times 100$

$\Rightarrow \frac{1}{4} = \frac{y - SP}{y}$

$= SP \text{ of } B = \frac{3y}{4}$

As SP of A = SP of B (Given)

$\frac{6x}{5} = \frac{3y}{4}$

$\Rightarrow \frac{6x}{5} - \frac{3y}{4} = 0$

$\Rightarrow 8x - 5y = 0$ ……..(ii)

By multiplying equ^{n}. (1) with 5 we get

$5x + 5y = 3250$ ……(iii)

Adding equ^{n}. (ii) & (iii)

$13x = 3250$

$x = 250$

So,

$250 + y = 650 \Rightarrow y = 650 - 250 = 400$

141. A fuse is used in mains electric supply as a safety device. Which one of the following statements about the fuse is correct?

(a) It is connected in parallel with the main switch
(b) It is made mainly from siver alloys
(c) It must have a low melting point
(d) It must have a very high resistance

Ans. (c) : As fuse is a safety device used in domestic electric circuit to prevent the house from short circuit. The fuse wire is made up of the metal having low melting point.

142. If 15 pumps of equal capacity can fill a tank in 7 days, then how many extra pumps will be required to fill the tank in 5 days?

(a) 6 (b) 7
(c) 14 (d) 21

Ans. (a) : No. of pumps required to fill a tank in 7 days = 15

$\therefore$ No. of pumps required to fill a tank in 1 day = 15 × 7 = 105 .....(i)

Let the extra pumps required be n to till the tant in 5 days.

Now, no of pumps required to fill the tank in 1 day = (n + 15) 5 .....(ii)

From (i) and (ii), (n + 15) 5 = 105

n + 15 = 21

n = 6

143. Out of the three annual examinations, each with a total of 500 mark, a student secured average marks of 45% and 55% in the first and second annual examinations. To have an overall average of 60%, how many marks does the student need to secure in the third annual examination?

(a) 450 (b) 400
(c) 350 (d) 300

Ans. (b) : Total marks for three examinations = 3 × 500 = 1500

Total required marks in three examinations = 60% of 1500 = (3 × 500 × 60)/100 = 900

Marks secured in first examination = 45 % of 500 = (500 × 45) / 100

=225

Marks secured in third examination = 55 % of 500

= (500 × 55) / 100 =275

Thus, the required marks in third examination =900 – (225 + 275) = 900 – 500

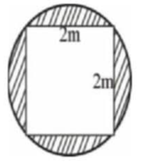
= 400

144. Which one of the following satisfies the relationship Dda : aDD : : Rrb : ?

(a) DDA (b) RRR
(c) bRR (d) BBR

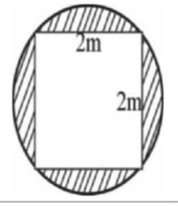
Ans. (c) : First and third letters have been interchanged. Middle letter becomes capital.

145. A square pond has 2 m sides and is 1 m deep. If it is to be enlarged the depth remaining the same, into a circular pond with the diagonal of the square as diameter as shown in the figure, then what would be the volume of earth to be removed?



(a) $(2\pi - 4)\text{m}^3$ (b) $(4\pi - 4)\text{m}^3$
(c) $(4\pi - 2)\text{m}^3$ (d) $(2\pi - 2)\text{m}^3$

Ans. (a) :



Diagonal of the square = $\sqrt{(2)^2 + (2)^2} = 2\sqrt{2}$

Radius of the circle = $\frac{2\sqrt{2}}{2} = \sqrt{2}$